Exam, Macroeconomics III

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Instructions: Answer all questions clearly and concisely. Show all your work where applicable. Explain intuition when asked.

Exercise 1 – Multiple Choice questions

Please answer the following multiple choice questions. For each question, write the question number and only the letter of the correct answer: "A", "B", "C", or "D". There is always exactly one correct answer. If you give more than one letter for a question, you receive zero points for that question. There are no negative points for wrong answers.

1. Consider an agent whose utility function is characterized by constant relative risk aversion (CRRA). This agent's optimal consumption path is described by the following Euler equation:

$$c_t^{-\sigma} = \beta(1+r)\mathbb{E}[c_{t+1}^{-\sigma}] \tag{1}$$

where $\sigma > 1$ is the inverse of the intertemporal elasticity of substitution, $0 < \beta < 1$ is the discount factor, $r > 0$ is the interest rate and c describes consumption. Consumption in period $t+1$ is uncertain, hence the agent forms expectations.

Suppose c_{t+1} becomes more uncertain, i.e., it's variance increases, but $\mathbb{E}[c_{t+1}]$ remains the same. What happens to c_t?

   A. c_t increases because higher future uncertainty raises expected marginal utility tomorrow.

   B. c_t decreases because higher uncertainty increases $\mathbb{E}[c_{t+1}^{-\sigma}]$, inducing precautionary saving.

   C. c_t remains unchanged because $\mathbb{E}[c_{t+1}]$ is unchanged.

   D. c_t decreases because higher uncertainty lowers the effective interest rate faced by the agent.

Correct answer: B) Uncertainty induces precautionary savings

2. Which of the following statements is an accurate description of the Beveridge Curve?

   A. The Beveridge Curve shows the relationship between output and unemployment over the business cycle.

   B. The Beveridge Curve describes the negative relationship between the unemployment rate and the vacancy rate in the labor market.

   C. The Beveridge Curve implies that higher worker bargaining power leads to fewer firm-worker matches.

   D. The Beveridge Curve represents the trade-off between inflation and unemployment implied by monetary policy.

   Correct answer: B) Relationship between v and u

3. Recall the Overlapping Generations Model from the lecture, in which agents live for two periods and only work in the first. In that model, which of the following statements is true?

   A. A pay-as-you-go pension system can never increase welfare.

   B. A fully funded pension system has no effect on the aggregate capital stock, as savers fully undo the forced pension savings through private borrowing or saving.

   C. It is irrelevant for welfare considerations whether there are infinitely many future generations or not.

   D. Ricardian equivalence holds, i.e., it does not matter for any agent's consumption whether a government that spends in period t recoups the money through taxes in the same period or ten periods later.

   Correct answer: B) A fully funded pension system has no effect on the aggregate capital stock

4. In the Ramsey growth model, economists often impose the Transversality Condition or the No-Ponzi Game condition. Why are these conditions important?

   A. They ensure that the interest rate stays constant in steady state.

   B. On a balanced growth path, as capital k increases, the real wage increases ($w = (1-\alpha)k^{\alpha}$). Without these conditions, hours worked would go to zero ($h \to 0$) due to the income effect.

   C. The conditions rule out explosive transition dynamics by preventing households from, e.g., accumulating too much debt.

   D. The conditions force consumption to grow at the same rate as output along the transition towards the steady state.

   Correct answer: C) Prevent explosive dynamics

5. Assume the central bank sets policy under discretion and agents form rational expectations. Relative to a credible commitment policy, what happens on average to inflation and output?

   A. Higher inflation, output unchanged.

   B. Higher inflation, higher output.

   C. Lower inflation, higher output.

   D. Lower inflation, output unchanged.

   Correct: A) Discretion creates an inflation bias: expectations adjust, so average output stays at the natural level while average inflation is higher.

6. The current account (CA) and net foreign assets (NFA) are defined in open-economy macro. Are these stocks or flows?

   A. Both are flows.

   B. Both are stocks.

   C. CA is a stock and NFA is a flow.

   D. CA is a flow and NFA is a stock.

   Correct: D) CA is a flow; NFA is a stock.

Exercise 2 – Households and taxes

Consider a two-period economy, $t = 0, 1$, with a unit mass of households. There is a single consumption good. Wages and interest rates are exogenous and given by w_0, w_1 and r, respectively; households receive income in both periods.

All households are identical and have preferences

$$u(c_0, c_1) = \log c_0 + \beta \log c_1,$$

where $0 < \beta < 1$ is the discount factor and c_t is consumption in period t. All households are subject to the following budget constraints:

$$\begin{aligned} c_0 + a_1 &= w_0 - \tau_0 \\ c_1 &= (1 + r)a_1 + w_1 - \tau_1, \end{aligned}$$

where a_1 is the household's savings, and τ_t is a lump-sum tax.

a) Derive the Euler equation for the household. If households become more patient, how does this affect the consumption choices they make? Give an economic explanation.

b) Combining the Euler equation with the budget constraints, find a mathematical expression for c_0 as a function of the model's parameters (i.e., w_0, w_1, β, r). If the interest rate rises, how does c_0 change. Give an economic explanation.

c) What is the agent's marginal propensity to consume out of changes in w_0?

The government chooses government spending $\{g_0, g_1\}$ and levies lump-sum taxes paid by the household. Like the household, the government can borrow and lend at rate r. The government budget constraint is

$$g_0 + \frac{g_1}{(1 + r)} = \tau_0 + \frac{\tau_1}{(1 + r)}.$$

d) Show that, with these assumptions, conditional on g_0 and g_1, it is irrelevant for their consumption decision when the government decides to tax households. Explain the economic intuition behind this result.

e) Now assume that the households live hand-to-mouth, i.e, they consume all of their after-tax labor income every period. Do you expect the result from the previous exercise to hold in this setup? Explain why or why not. (You don't need to show it mathematically)

Exercise 3 – Flexible and rigid prices in the economy

Consider the following model as in the lectures. Assume a continuum of monopolistically competitive households of measure one. Goods are imperfect substitutes with elasticity of substitution $\theta > 1$. Aggregate demand is

$$Y = \frac{M}{P}, \tag{2}$$

where M is the money supply controlled by the central bank. Agents form rational expectations. Take as given following optimal price the household would like to charge

$$\frac{P_i}{P} = \frac{\theta}{\theta - 1} Y_i^{\phi - 1} \qquad \text{with } \phi > 1. \tag{3}$$

P_i and Y_i are the price and quantity of the good produced by consumer i, respectively.

a) How does the optimal price the household would like to charge in Equation (2) compare to a hypothetical economy in which prices are determined by perfect competition? Explain in words.

b) Show that, after taking logarithms, the average desired price can be written as

$$p^* = v + \gamma m + (1 - \gamma)p. \tag{4}$$

Small letters represent logs, such that: $\ln(X) = x$. Find an expression for γ as a function of the model parameters. Under what assumptions is γ large? Give an economic reason for your answer. (Hint: assume that the aggregate variable p^* is the averages of the individual household variables p_i. Further, since all households are identical, $y = y_i$.)

For the remainder of the exercise, we will suppress the constant v – you may include it in your solutions, but, for brevity, you don't need to. Now, assume that a fraction δ of households can set their prices perfectly flexibly. Their price level is given by

$$p^f = \gamma m + (1 - \gamma)p. \tag{5}$$

The remaining households' prices are more rigid, and are set before the money supply is realized. Their prices are given by

$$p^r = \gamma \mathbb{E}[m] + (1 - \gamma)\mathbb{E}[p]. \tag{6}$$

Thus, the overall price level in the economy is $p = \delta p^f + (1-\delta)p^r$. Since this is a problem of sequential price setting (flexible price-setters set their prices last), it is possible to solve for the model's solution by backward induction.

c) Find an expression for p^f, given p^r, m and the model's parameters. In the hypothetical case of $\delta \to 1$, what would be the price level in the economy? Explain.

d) Show that the the rigid price households set their prices such that $p^r = \mathbb{E}[m]$. What is the economic intuition behind this result?

e) Show that the equilibrium values of the price level and output are given by

$$p = \mathbb{E}[m] + \frac{\delta\gamma}{1-(1-\gamma)\delta}\left(m - \mathbb{E}[m]\right) \tag{7}$$

$$y = \frac{1-\delta}{1-(1-\gamma)\delta}\left(m - \mathbb{E}[m]\right). \tag{8}$$

Explain how the price influences the transmission of an unexpected change in the money supply m. Discuss the validity of the statement "only unexpected changes in the money supply affect the real economy".

Suggested solutions

Exercise 2

a) Combine constraints into the intertemporal budget:

$$c_0 + \frac{c_1}{1+r} = (w_0 - \tau_0) + \frac{w_1 - \tau_1}{1+r}.$$

First-order conditions give:

$$\begin{aligned} &\frac{1}{c_0} = \lambda, \\ &\frac{\beta}{c_1} = \frac{\lambda}{1+r} \quad \Rightarrow \quad c_1 = \beta(1+r)c_0. \end{aligned}$$

Interpretation: Higher β (more patient) $\Rightarrow$ consume less today and more tomorrow.

b) Substitute Euler equation into budget constraint:

$$\begin{aligned} c_0 + \frac{c_1}{1+r} &= c_0 + \frac{\beta(1+r)c_0}{1+r} \\ &= c_0(1+\beta) \\ &= (w_0 - \tau_0) + \frac{w_1 - \tau_1}{1+r}. \end{aligned}$$

$$c_0 = \frac{(w_0 - \tau_0) + \frac{w_1 - \tau_1}{1+r}}{1+\beta}$$

Effect of r: As $r \uparrow$, $\frac{w_1-\tau_1}{1+r} \downarrow \Rightarrow c_0$ decreases. This is a *wealth effect*: the present value of future resources falls. With log utility ($\sigma = 1$), substitution and income effects cancel.

c)

$$\frac{\partial c_0}{\partial w_0} = \frac{1}{1+\beta}.$$

d) From above, we have

$$c_0 = \frac{(w_0 - \tau_0) + \frac{w_1 - \tau_1}{1+r}}{1+\beta}$$

Substituting in the government's budget constraint, this becomes

$$c_0 = \frac{(w_0 - g_0) + \frac{w_1 - g_1}{1+r}}{1+\beta}$$

Since taxes do not show up, they must be irrelevant for the consumption choice. Only the *present value of taxes* matters, which is given by the path of government spending. Timing is irrelevant. Households save to offset future taxes $\Rightarrow$ debt-financed fiscal policy has no effect on consumption.

e) If $c_t = w_t - \tau_t$, households cannot save. Ricardian equivalence fails: temporary tax cuts increase c_0, so fiscal stimulus is effective.

Exercise 3

a) *Perfect competition:* as $\theta \to \infty$, the markup $\frac{\theta}{\theta-1} \to 1$, so

$$\frac{P_i}{P} = Y_i^{\varphi-1}.$$

b) Taking logs of $\frac{P_i}{P} = \frac{\theta}{\theta-1} Y_i^{\varphi-1}$ gives

$$p_i = v + (\varphi - 1)y_i + p, \qquad v \equiv \ln\left(\frac{\theta}{\theta - 1}\right).$$

In a symmetric equilibrium $y_i = y$ and from $Y = M/P$ we have $y = m - p$. Hence

$$p^* - p = v + (\varphi - 1)(m - p),$$

so

$$p^* = v + (\varphi - 1)m + (2 - \varphi)p.$$

Comparing to $p^* = v + \gamma m + (1 - \gamma)p$ yields

$$\boxed{\gamma = \varphi - 1.}$$

γ is large when φ is large (labor supply is more inelastic / marginal cost rises steeply with output), so nominal demand changes translate more into prices and less into quantities.

c) With flexible setters:

$$p^f = \gamma m + (1 - \gamma)p, \qquad p = \delta p^f + (1 - \delta)p^r.$$

Substitute p into the flexible rule:

$$p^f = \gamma m + (1 - \gamma)\big(\delta p^f + (1 - \delta)p^r\big),$$

so

$$p^f\big(1 - (1 - \gamma)\delta\big) = \gamma m + (1 - \gamma)(1 - \delta)p^r,$$

and therefore

$$\boxed{p^f = \frac{\gamma}{1 - (1 - \gamma)\delta}\, m + \frac{(1 - \gamma)(1 - \delta)}{1 - (1 - \gamma)\delta}\, p^r.}$$

If $\delta \to 1$, then all prices are flexible and the price level satisfies $p = m$.

d)

$$
\begin{aligned}
p^r &= \gamma\mathbb{E}[m] + (1-\gamma)\,\mathbb{E}\big[\delta p^f + (1-\delta)p^r\big] \\
p^r &= \gamma\mathbb{E}[m] + (1-\gamma)\,\mathbb{E}\left[\delta\left(\frac{\gamma}{1-(1-\gamma)\delta}m + \frac{(1-\gamma)(1-\delta)}{1-(1-\gamma)\delta}p^r\right) + (1-\delta)p^r\right] \\
p^r\big(1-(1-\gamma)\delta\big) &= (1-(1-\gamma)\delta)\gamma\mathbb{E}[m] + (1-\gamma)\delta\gamma\mathbb{E}[m] \\
&\quad + (1-\gamma)\delta(1-\gamma)(1-\delta)\mathbb{E}[p^r] + (1-\gamma)(1-\delta)\big(1-(1-\gamma)\delta\big)\mathbb{E}[p^r] \\
p^r\big(1-(1-\gamma)\delta\big) &= \gamma\mathbb{E}[m] + (1-\gamma)(1-\delta)\mathbb{E}[p^r] \\
p^r &= \mathbb{E}[m].
\end{aligned}
$$

Since the response of the flexible price setters will be optimal, and flexible price setters can set their prices last, rigid price setters cannot exploit the economic order. They set their prices such that they match their best guess for the optimal prices set in period t.

e) Using $p = \delta p^f + (1-\delta)p^r$ and the expression from (d), with $p^r = \mathbb{E}[m]$:

$$
p = \delta\left(\frac{\gamma}{1-(1-\gamma)\delta}\,m + \frac{(1-\gamma)(1-\delta)}{1-(1-\gamma)\delta}\,\mathbb{E}[m]\right) + (1-\delta)\mathbb{E}[m]
$$

which simplifies to

$$
\boxed{p = \mathbb{E}[m] + \frac{\delta\gamma}{1-(1-\gamma)\delta}\big(m - \mathbb{E}[m]\big).}
$$

Since $y = m - p$,

$$
\boxed{y = \frac{1-\delta}{1-(1-\gamma)\delta}\big(m - \mathbb{E}[m]\big).}
$$

More rigidity (smaller δ) means surprises in m feed more into y and less into p; anticipated changes are priced in and do not move y.